

Unramifiedness of $\mathcal{P}^{[d]G}$ at the exceptional divisor

(Herbrand-convexity route, for comparison)

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Scope. This note gives a proof of ?? (= `thm:main` of `proof_complete_standalone.tex`) routed through local ramification theory. The new content is a convexity lemma for the Herbrand φ -function on a cyclic p -group (??). The route works only if one accepts the local–global dictionary stated in ?? below as a separately-established result; we discuss the resulting circularity at the end (??).

We adopt the notation of `proof_complete_standalone.tex` without restating the background: C is a smooth projective curve over k , $p_\infty \in C$, $U = C \setminus \{p_\infty, \dots\}$, $\mathcal{P} \rightarrow U$ a G -torsor with $G \cong \mathbb{Z}/p^r\mathbb{Z}$ a cyclic p -group, $H \subseteq G$ a subgroup of order p^s , $f: C' \rightarrow C$ the compactification of the G/H -torsor $\mathcal{P}_{G/H} \rightarrow U$, $p'_\infty \in f^{-1}(p_\infty)$, and $X_C, X_{C'}$ the blowups of $C^{(d)}, C'^{(d)}$ at $d \cdot p_\infty, d \cdot p'_\infty$ with exceptional generic points $\eta_C, \eta_{C'}$. Throughout, $0 < s < r$ (the case $s = 0$ or $s = r$ being trivial).

1 The Herbrand convexity lemma

The single new local ingredient. We work in the setting of a finite totally ramified Galois extension of complete (equivalently, henselian and defectless) discrete valuation fields with cyclic p -group Galois group. Lower and upper numbering for ramification follow [Serre1979], Ch. IV.

Lemma 1. *Let $\mathcal{F}/\mathcal{K}$ be a finite totally ramified Galois extension of complete discrete valuation fields with Galois group $\Gamma \cong \mathbb{Z}/p^r\mathbb{Z}$, $r \geq 1$, and let $\Delta \subseteq \Gamma$ be the unique subgroup of order p^s with $0 < s < r$, so that $\Gamma/\Delta \cong \mathbb{Z}/p^{r-s}\mathbb{Z}$ and $|\Delta| = p^s$. Fix an integer $d \geq 1$. If both*

- (a) $(\Gamma/\Delta)^d = \{1\}$, i.e. the upper ramification filtration of Γ/Δ acting on $\mathcal{F}^\Delta/\mathcal{K}$ is trivial in degree $\geq d$, and
- (b) $\Delta^d = \{1\}$, i.e. the upper ramification filtration of Δ acting on $\mathcal{F}/\mathcal{F}^\Delta$ is trivial in degree $\geq d$,

then $\Gamma^d = \{1\}$, i.e. the upper ramification filtration of Γ acting on $\mathcal{F}/\mathcal{K}$ is trivial in degree $\geq d$.

Proof. Let

$$\Gamma = \Gamma_0 \supsetneq \Gamma_{m_1+1} \supsetneq \Gamma_{m_2+1} \supsetneq \dots \supsetneq \Gamma_{m_r+1} = \{1\}$$

be the lower-numbering ramification filtration of Γ on $\mathcal{F}/\mathcal{K}$, with jumps at integers $m_1 < m_2 < \dots < m_r$. Since $|\Gamma| = p^r$ and the extension is totally ramified, there are exactly r jumps, and $|\Gamma_{m_j+1}| = p^{r-j}$. Set

$$\Delta_0 := m_1, \quad \Delta_j := m_{j+1} - m_j \quad (1 \leq j \leq r-1),$$

so that $m_j = \Delta_0 + \Delta_1 + \dots + \Delta_{j-1}$. (We reuse the letter Δ for the gap sequence; the subgroup $\Delta \subseteq \Gamma$ will appear only as such, with subscripts to differentiate.)

The three Herbrand functions. Recall $\varphi_\Gamma(u) = \int_0^u |\Gamma_t|/|\Gamma_0| dt$. The integrand takes the constant value p^{-j} on $(m_j, m_{j+1}]$ ($0 \leq j \leq r-1$, with $m_0 := 0$), so

$$\varphi_\Gamma(m_r) = \sum_{j=0}^{r-1} p^{-j} \Delta_j. \quad (1)$$

For the subgroup $\Delta = \Gamma_{m_{r-s}+1}$ acting on $\mathcal{F}/\mathcal{F}^\Delta$, the lower ramification of Δ as a subgroup of Γ is given by $\Delta_i = \Delta \cap \Gamma_i$ ([Serre1979], Ch. IV, §1, Prop. 2). Therefore $\Delta_i = \Delta$ for $0 \leq i \leq m_{r-s}$, $\Delta_i = \Gamma_i$ for $i > m_{r-s}$, and the lower jumps of Δ occur precisely at $m_{r-s+1} < m_{r-s+2} < \dots < m_r$. Hence

$$\varphi_\Delta(m_r) = m_{r-s+1} + \sum_{k=1}^{s-1} p^{-k} (m_{r-s+k+1} - m_{r-s+k}) = m_{r-s} + \sum_{k=0}^{s-1} p^{-k} \Delta_{r-s+k}. \quad (2)$$

For the quotient Γ/Δ acting on $\mathcal{F}^\Delta/\mathcal{K}$, Herbrand's theorem ([Serre1979], Ch. IV, §3, Lemma 5) gives $(\Gamma/\Delta)^v = \Gamma^v \Delta/\Delta$. Writing $u_j := \varphi_\Gamma(m_j)$ for the upper jumps of Γ , one has $\Gamma^{u_j} = \Gamma_{m_j}$, so $\Gamma^{u_j} \Delta \supsetneq \Delta$ iff $\Gamma_{m_j} \not\subseteq \Delta$, iff $j \leq r-s$. The highest upper jump of Γ/Δ is therefore $u_{r-s} = \varphi_\Gamma(m_{r-s})$, and

$$u_{r-s} = \varphi_\Gamma(m_{r-s}) = \sum_{j=0}^{r-s-1} p^{-j} \Delta_j. \quad (3)$$

The fundamental identity. Split the sum in equation (??) at index $j = r-s$ and factor $p^{-(r-s)}$ from the tail:

$$\varphi_\Gamma(m_r) = \underbrace{\sum_{j=0}^{r-s-1} p^{-j} \Delta_j}_{=u_{r-s} \text{ by equation (??)}} + p^{-(r-s)} \underbrace{\sum_{k=0}^{s-1} p^{-k} \Delta_{r-s+k}}_{=\varphi_\Delta(m_r) - m_{r-s} \text{ by equation (??)}}.$$

Hence

$$\boxed{\varphi_\Gamma(m_r) = u_{r-s} + \frac{\varphi_\Delta(m_r) - m_{r-s}}{p^{r-s}}.} \quad (4)$$

Convex combination. Set $A := u_{r-s}$ and $B := \varphi_\Delta(m_r)$. Hypotheses (a) and (b) read $A < d$ and $B < d$ respectively, by the characterisation of the highest upper jump as the smallest v such that the upper group is trivial in degree $> v$. Since the integrand defining φ_Γ is bounded above by

1, we have $\varphi_\Gamma(m_{r-s}) \leq m_{r-s}$, i.e. $A \leq m_{r-s}$. Substituting into equation (??), $B - m_{r-s} \leq B - A$, hence

$$\varphi_\Gamma(m_r) \leq A + \frac{B-A}{p^{r-s}} = \left(1 - \frac{1}{p^{r-s}}\right)A + \frac{1}{p^{r-s}}B.$$

The right-hand side is a convex combination of two quantities both strictly less than d , hence

$$\varphi_\Gamma(m_r) < \left(1 - \frac{1}{p^{r-s}}\right)d + \frac{1}{p^{r-s}}d = d.$$

The highest upper jump of Γ is $\varphi_\Gamma(m_r) < d$, so $\Gamma^d = \{1\}$, as claimed. $\square$

The cyclic prime-power hypothesis is essential: it forces both the chain of lower-numbering ramification groups to pass through every subgroup of Γ (which lets us identify the lower jumps of Δ as a tail of the jumps of Γ) and the existence of a unique Δ of each order (so the hypotheses (a), (b) make unambiguous sense). For a non-cyclic abelian p -group, the analogous statement fails; the convexity step still works but the lower jumps of $\Delta \subseteq \Gamma$ need not be a tail.

2 The dictionary, as a black box

The geometric-to-local translation we need takes the following form.

[Local-global dictionary for symmetric-power torsors] Let Z be a smooth projective geometrically connected curve over a field k , $q \in Z$ a closed point, Γ a finite abelian group, and $\mathcal{Q} \rightarrow Z \setminus \{q\}$ a Γ -torsor with associated complete local extension $\mathcal{F}/\mathcal{K}_q$ at q (in the sense of ?? of `proof_complete_standalone.tex`). Let $\pi_Z: X_Z \rightarrow Z^{(d)}$ be the blowup at $d \cdot q$, with exceptional divisor of generic point η_Z . Then

$$\mathcal{Q}^{[d]_\Gamma} \rightarrow (Z \setminus \{q\})^{(d)} \text{ is unramified at } \eta_Z \iff \Gamma^d = \{1\} \text{ on } \mathcal{F}/\mathcal{K}_q.$$

We will use this proposition three times in the proof of ??. We discuss its status in ??.

3 The proof

Theorem 2. *Suppose*

- (H1) $(\mathcal{P}_{G/H})^{[d]} \rightarrow U^{(d)}$ is unramified at η_C ;
- (H2) $\mathcal{P}^{[d]_H} \rightarrow (\mathcal{P}_{G/H})^{(d)}$ is unramified at $\eta_{C'}$;
- (H3) $G = \mathbb{Z}/p^r\mathbb{Z}$ is cyclic of prime-power order.

Then $\mathcal{P}^{[d]_G} \rightarrow U^{(d)}$ is unramified at η_C .

Proof. After the standard reductions (replacing G by the image of monodromy and H by its intersection; we may assume $\mathcal{P}$ is a connected G -torsor and that $0 < |H| < |G|$), we work with the complete local extension $\mathcal{F}/\widehat{\mathcal{K}}_{p_\infty}$ corresponding to $\mathcal{P} \rightarrow U$ at p_∞ , which is totally ramified with Galois group G (after passing to a connected component of $\mathcal{P}|_{\text{Spec } \widehat{\mathcal{K}}_{p_\infty}}$ and extending scalars if needed). The subextension $\mathcal{F}^H \subseteq \mathcal{F}$ has $\text{Gal}(\mathcal{F}^H/\widehat{\mathcal{K}}_{p_\infty}) = G/H$ and $\text{Gal}(\mathcal{F}/\mathcal{F}^H) = H$. The choice of $p'_\infty \in f^{-1}(p_\infty)$ corresponds to choosing a place of $\mathcal{F}^H$, with completion identified with $\widehat{\mathcal{K}}_{p'_\infty}$; the local extension of the H -torsor $\mathcal{P} \rightarrow \mathcal{P}_{G/H}$ at p'_∞ is then $\mathcal{F}/\mathcal{F}^H$.

Step 1: From (H1) to a local jump bound for G/H . Apply ?? to the (G/H) -torsor $\mathcal{P}_{G/H} \rightarrow U$ at p_∞ , with $\Gamma = G/H$. Hypothesis (H1) is the geometric side of the equivalence, so the local side gives

$$(G/H)^d = \{1\} \text{ on } \mathcal{F}^H / \widehat{\mathcal{K}}_{p_\infty}.$$

Step 2: From (H2) to a local jump bound for H . Apply ?? to the H -torsor $\mathcal{P} \rightarrow \mathcal{P}_{G/H}$ at the closed point $p'_\infty \in C'$, with base curve $Z = C'$, group $\Gamma = H$, and the same integer d . The blowup of $C'^{(d)}$ at $d \cdot p'_\infty$ is exactly $X_{C'}$ of our master diagram, with exceptional generic point $\eta_{C'}$; the geometric side of the equivalence is therefore (H2). The local side reads

$$H^d = \{1\} \text{ on } \mathcal{F} / \mathcal{F}^H.$$

Step 3: Apply the Herbrand convexity lemma. ?? applies with $\mathcal{K} = \widehat{\mathcal{K}}_{p_\infty}$, $\mathcal{F} = \mathcal{F}$, $\Gamma = G$, $\Delta = H$: hypotheses (a) and (b) are Steps 1 and 2 respectively, and (H3) is the cyclic prime-power assumption. We conclude

$$G^d = \{1\} \text{ on } \mathcal{F} / \widehat{\mathcal{K}}_{p_\infty}.$$

Step 4: Translate back via the dictionary. Apply ?? once more, now to the G -torsor $\mathcal{P} \rightarrow U$ at p_∞ with $\Gamma = G$. The local side is Step 3; the geometric side reads

$$\mathcal{P}^{[d]G} \rightarrow U^{(d)} \text{ is unramified at } \eta_C,$$

which is the conclusion of ??. □

4 Discussion: where this proof is circular, and how to rescue it

[Circularity in Step 4] The application of ?? in Step 4 of the proof above is for the group G itself — the same group that appears in the conclusion of ??. If ?? is read as a known theorem established elsewhere (say in the cited literature for sufficiently general abelian Γ), this is fine; but if the dictionary is itself the subject of the work and ?? is one of its building blocks, then Step 4 is circular.

The standard way to rescue the argument is to embed it in an *induction on $|G|$* . The dictionary for cyclic p -groups of order $< p^r$ is then the inductive hypothesis, Steps 1 and 2 use the dictionary in the easy direction for the strictly smaller groups G/H and H , and Step 4 establishes the hard direction of the dictionary for G . This is presumably what the original draft `qwn_atm_1.txt` intended; making the induction explicit (including the base case $r = 1$, the trivial subgroup cases handled by the reductions, and a check that the smaller groups $G/H \cong \mathbb{Z}/p^{r-s}\mathbb{Z}$ and $H \cong \mathbb{Z}/p^s\mathbb{Z}$ are both of order $< p^r$ when $0 < s < r$) is what is required to make the present proof rigorous.

[Comparison with the direct Cartesian-inertia proof] The proof of the same theorem in `claudé_atm_1.tex` avoids ?? entirely. It works directly with inertia groups in the function-field tower $K_C \subseteq F \subseteq E$ inside its compositum with $K_{C'}$, uses the Cartesian property of the master diagram to relate $I_{v_E}(E/F) \cap N$ to the trivially-inertia subgroup $I_{v_*}(E \cdot K_{C'} / K_{C'})$ (which is trivial by (H2)), and then deploys (H3) via the chain-of-subgroups property of cyclic p -groups to conclude $I_{v_E}(E/F) = 1$.

That proof has three advantages over the present one:

- It uses no local ramification theory: no Herbrand function, no upper numbering, no induction on $|G|$.
- It does not invoke the dictionary at all, so the circularity issue of ?? does not arise.
- The role of (H3) is transparent — it is the chain property of cyclic p -groups — whereas the present proof loads (H3) into both ?? (via the chain of lower-numbering ramification groups) and the dictionary (via whatever proof of ?? one cites).

The present proof's only structural advantage is that ?? is a clean, self-contained statement of local ramification theory that may be of independent interest, and that the proof factors cleanly through the (well-studied) local–global dictionary rather than through an ad-hoc Galois argument inside a particular tower. For thesis exposition the direct proof seems preferable; the Herbrand-route proof is better seen as motivation for the lemma than as the primary proof.